

JULIANA (JULI) STANLEY
Ph.D. Student
Massachusetts Institute of Technology

A. PERSONAL

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B. EDUCATION

09/2020-Present **Ph.D. (ongoing)**, Department of Biology
Massachusetts Institute of Technology
Advisor: Dr. Gene-Wei Li

09/2018-05/2020 **M.S.** Bioinformatics, College of Science
Northeastern University

09/2015-12/2019 **B.S.** Biology, Department of Biology
Northeastern University
Advisor: Javier Apfeld
Thesis: *"Sensitivity analysis of two-state ratiometric biosensors"*

C. RESEARCH POSITIONS HELD

05/2021-Present **Ph.D. Student, Laboratory of Dr. Gene-Wei Li**
Massachusetts Institute of Technology
Characterizing mechanisms of leaderless mRNA translation in bacteria

05/2020-09/2020 **Software Developer, R Project for Statistical Computing**
Google Summer of Code
Developing an interactive interface for HMM design to identify change points

- 07/2019-05/2020** **Visiting M.S. Student, Laboratory of Dr. Pamela Silver**
Harvard Medical School
Engineering proteins with functions robust to freezing and oxidation
- 10/2015-07/2019** **Undergraduate Student, Laboratory of Dr. Javier Apfeld**
Northeastern University
Characterizing oxidative stress tolerance in *C. elegans*
- 05/2018-09/2018** **Undergraduate Student, Laboratory of Dr. Jeremy Gunawardena**
Harvard Medical School
Subtyping clinical tumor samples from genomic information
- 07/2017-05/2018** **Research Associate, Immunology**
Cygnal Therapeutics
Identifying modulators of macrophage polarization

D. TEACHING POSITIONS HELD

- 02/2024-05/2024** **Graduate Teaching Assistant for 7.571 Quantitative Analysis of Biological Data and 7.572 Quantitative Measurements and Modelling of Biological Systems**
Massachusetts Institute of Technology
Led discussion and problem-solving recitations (2/week) and graded in a graduate quantitative biology course.
- 09/2021-12/2021** **Graduate Teaching Assistant for 7.015 Introductory Biology**
Massachusetts Institute of Technology
Led discussion and problem-solving recitations (2/week) and graded in an undergraduate introductory biology course.
- 06/2020-09/2020** **Community Teaching Assistant for 7.28.1x-7.28.3x**
edX
Answered questions from a global community of students for MIT's molecular biology MOOCs.
- 09/2019-05/2020** **Graduate Teaching Assistant for Bioinformatics Computational Methods I/II**
Northeastern University
Led recitations, served as a substitute lecturer, developed automated grading scripts, and refined parts of the core curriculum.
- 09/2016-12/2016** **Research Assistant, Laboratory for Genetics and Molecular Biology**
Northeastern University
Developed hands-on computational genetics teaching modules.

- 05/2016-09/2016 Undergraduate Teaching Assistant**
Northeastern University
Facilitated student groups developing independent, techniques-focused laboratory projects.
- 01/2016-05/2016 Research Assistant, Laboratory for General Chemistry I**
Northeastern University
Helped to develop and refine a project-based introductory chemistry course.

E. AWARDS, FELLOWSHIPS, AND RESEARCH GRANTS

- 01/2019-05/2019 Honors Interdisciplinary Thesis Grant**
Northeastern University
- 05/2019-09/2019 Scholars Independent Research Fellowship (Proposal-based fellowship)**
05/2017-09/2017 Northeastern University
05/2016-09/2016
- 01/2017-05/2017 Honors Undergraduate Research Grant (Proposal-based project funding)**
01/2016-05/2016 Northeastern University
- 09/2016-12/2016 Provost Undergraduate Research Grant (Proposal-based project funding)**
Northeastern University
- 09/2015-05/2020 University Scholars Program (Highest Merit Scholarship)**
Northeastern University

F. PUBLICATIONS, PREPRINTS AND PATENTS

Previous name for publication: Julian A. Stanley. See ORCID for published aliases.

Publications

1. Herzel, L.; **Stanley, J. A.**; Yao, C.-C.; & Li, G.-W. Ubiquitous mRNA decay fragments in *E. coli* redefine the functional transcriptome. *Nucleic Acids Research* **2022**. <https://doi.org/10.1093/nar/gkac295>
2. **Stanley, J. A.**; Johnsen, S. B.; Apfeld, J. The SensorOverlord Predicts the Accuracy of Measurements with Ratiometric Biosensors. *Sci Rep* **2020**, 10 (1), 16843. <https://doi.org/10.1038/s41598-020-73987-0>.
3. Schiffer, J. A.; Servello, F. A.; Heath, W. R.; Amrit, F. R. G.; Stumbur, S. V.; Eder, M.; Martin, O. M.; Johnsen, S. B.; **Stanley, J. A.**; Tam, H.; Brennan, S. J.; McGowan, N. G.; Vogelaar, A. L.; Xu, Y.; Serkin, W. T.; Ghazi, A.; Stroustrup, N.; Apfeld, J. Caenorhabditis Elegans Processes Sensory Information to Choose between Freeloading and Self-Defense Strategies. *eLife* **2020**, 9, e56186. <https://doi.org/10.7554/eLife.56186>.

4. Chang, R. L.; **Stanley, J. A.**; Robinson, M. C.; Sher, J. W.; Li, Z.; Chan, Y. A.; Omdahl, A. R.; Wattiez, R.; Godzik, A.; Matallana-Surget, S. Protein Structure, Amino Acid Composition and Sequence Determine Proteome Vulnerability to Oxidation-induced Damage. *The EMBO Journal* **2020**, 39 (23). <https://doi.org/10.15252/embj.2020104523>.

Teaching-Related Publications

1. Yang, Kevin and **Stanley, J.** Predicting smoking status based on RNA sequencing data. *Journal of Emerging Investigators* **2024**, Vol 7. <https://doi.org/10.59720/23-099>.

Preprints

1. de la Parra, J.; **Stanley, J. A.**; Foster, S.; Webb, C.; Lykourinou, V. Crafting A More Environmentally Benign Extraction and Analysis of Pharmaceutical Precursors from a Medicinal Plant: A Student-Led Innovation; preprint; *chemrxiv*, 2020. <https://doi.org/10.26434/chemrxiv.7791716.v1>.

Patents

1. Kahvejian, A.; Mata-Fink, J.; HUROV, J. B.; SHU, C. J.; NEUBAUER, G. H.; FANKHAUSER, M. A.; **STANLEY, J. A.** Methods and Compositions for Treating Inflammatory or Autoimmune Diseases or Conditions Using Serotonin Receptor Activators. US11208475B1, December 28, 2021.
2. Kahvejian, A.; Mata-Fink, J.; HUROV, J. B.; SHU, C. J.; NEUBAUER, G. H.; **STANLEY, J. A.** Methods and Compositions for Treating Inflammatory or Autoimmune Diseases or Conditions Using GRM8 Activators. US11059886B1, July 13, 2021.
3. Kahvejian, A.; Mata-Fink, J.; HUROV, J. B.; SHU, C. J.; NEUBAUER, G. H.; **STANLEY, J. A.** Methods for Treating Cancer Using GRM8 Inhibitors. US10683352B1, June 16, 2020.
4. Kahvejian, A.; Mata-Fink, J.; HUROV, J. B.; SHU, C. J.; NEUBAUER, G. H.; FANKHAUSER, M. A.; **STANLEY, J. A.** Methods and Compositions for Treating Cancer Using Serotonin Receptor Inhibitors. US11034751B1, June 15, 2021.

G. SERVICE AND MENTORSHIP

2023, 2025

MIT Quantitative Methods Workshop

Instructor for a 4-hour quantitative RNA sequencing workshop for undergraduates with limited access to research experience at their home institution.

2022-2023

Polygence (Remote)

Bioinformatics project mentor, guiding high school students in independent research projects with clearly defined checkpoints and goals.

- 2021-2024** **MIT Biology Application Assistance Program (BAAP)**
Founded BAAP to support students applying to biology Ph.D. programs. Helped to organize panels, mentor pairings, and information sessions with 50+ graduate students and 100+ applicants. After 3 years, facilitated a smooth transition of program leadership to earlier-stage graduate students.
- 2020, 2021** **Científico Latino**
Participated as a mentor to Ph.D. applicants from backgrounds historically underrepresented in biology. Advised:
- Sefiyat Adebisi (2021, currently: incoming Ph.D. Student, Cornell University)
 - Kaylie Pinto (2020 currently: Ph.D. Student, University of Maryland, Baltimore)
- 2020, 2021** **CovEducation**
Mentored a high school student in history, chemistry, computer science, and college preparation from the start of the COVID-19 pandemic until entering college. Advised:
- Eileen Ye (2020-2021, currently: undergraduate, Harvard University)
- 2020, 2021, 2022** **Boston Public Schools Science Fair**
Science fair judge for Region V and VI middle and high school students in biology and computer science.
- 2016-2019** **Northeastern University Biochemistry Club**
Wrote organization grants, including applying and implementing two years of the Northeast-regional undergraduate conference of The American Society of Biochemistry and Molecular Biology.
- 2015-2017** **Peer Health Exchange**
Taught weekly, 30–45-minute classes about physical and emotional health to high school students in Boston area public schools.